

AI Extraction of Disassembly Videos for Generating AR Repair Instructions

Wenjing Deng
Tsinghua University
Beijing, China
dengwj24@mails.tsinghua.edu.cn

Yiran Wan
Tongji University
Shanghai, China
3246606388@qq.com

Sisi He
Nanyang Institute of Technology
Nanyang, China
2668853005@qq.com

Chenzhuo Xiang
Tsinghua University
Beijing, China
xcz24@mails.tsinghua.edu.cn

Xintong Wu
Tsinghua University
Beijing, China
wuxt22@mails.tsinghua.edu.cn

Zhihao Yao
Tsinghua University
Beijing, Beijing, China
yaozh_h@outlook.com

Haipeng Mi
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

Abstract

This paper introduces a lightweight AI-assisted framework that converts disassembly videos into interactive augmented reality (AR) repair guidance, supporting novice users who often struggle with part organization and reassembly sequencing during household repairs. Our system integrates vision-language models in a multi-agent pipeline to automatically extract procedural steps, component identities, and spatial relationships from user-recorded videos during disassembly, enabling it to support diverse products while reducing AR authoring effort. In an initial evaluation with six products, the system correctly identified 29 of 32 components and produced accurate task descriptions, showing its potential for cross-product repair support.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

Keywords

Repair guidance, Augmented reality

ACM Reference Format:

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1 Introduction

Augmented reality (AR) is increasingly used in repair and maintenance tasks [3, 5–7, 9] by overlaying real-time visual guidance onto physical environments. In household repair scenarios, prior studies have shown that AR can reduce cognitive load [7] and improve usability and user satisfaction compared to traditional video tutorials [5]. However, most existing AR guidance systems are built for a specific product and rely on 3D overlays [6] to convey procedural and spatial information, making them costly to develop and difficult to scale across products.

Advances in computer vision and large language models (LLMs) now make it possible to extract procedural knowledge directly from videos [2, 8, 10]. Despite these advances, most systems focus on task segmentation or narrative structure, with little consideration for the spatial or mechanical relationships between components, which are critical for guiding accurate actions in physical tasks.

To address these gaps, we present an AI-assisted framework that generates interactive AR reassembly guidance for household appliances using user-recorded disassembly videos as input. The system requires no prior product models or 3D authoring. It employs vision-language models to identify components, infer spatial connections (e.g., A fixed to B), and produce reverse reassembly steps. This information is then visualized through interactive 2D exploded diagrams paired with reversed disassembly video clips, providing spatial clarity while keeping AR authoring effort low.

2 User Interface Design

To inform our design, we first observed experienced repairers performing disassembly tasks, and then conducted teardown workshops with novice users to identify their challenges. From these observations, we identified three types of information as essential for effective guidance: the sequence of steps, the components involved in each step, and the spatial alignment and connections between components. This insight echoes findings from prior work on assembly and procedural tasks [4], which also highlight the importance of these elements. Guided by these findings, we designed the following features and visualizations:

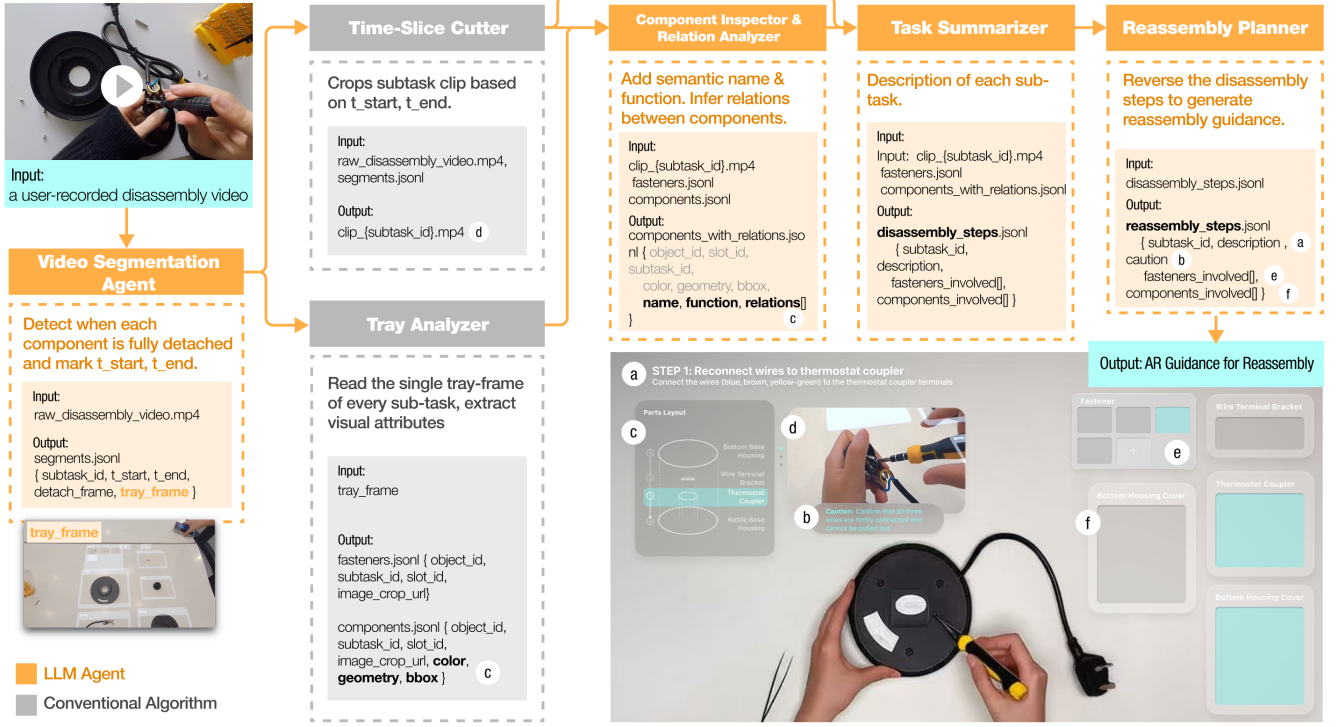


Figure 1: The system segments the input video into repair steps, processes each subtask with VLLMs using custom-designed prompts to extract key details, and visualizes the resulting instructions in the AR interface.

Task Flow Overview: Provides users with a clear overview of the repair process, outlining the sequence of sub-steps and highlighting the user’s current step (Figure 1.a).

Component Identification Guidance: Assists users in organizing components during disassembly with virtual trays (Figure 1.e & f). During reassembly, the highlighted tray indicates the components and fasteners required for each step.

Component Alignment Guidance: Provides detailed instructions on how components should align and fit together. Instead of relying on 3D assets, we use exploded diagrams (Figure 1.c) accompanied by reverse-edited video clips to convey alignment information. Users can also click on components in the diagram to jump to the corresponding video clip.

3 System Overview and Implementation

We developed a multi-agent framework to automatically generate AR reassembly guidance from user-recorded disassembly videos: **Video Segmentation Agent** detects the moment of each key component is detached as we define each subtask as the removal of a key component, and records four key timestamps: the start (t_{start}) and end (t_{end}) of the subtask, the frame where the component is fully detached ($detach_frame$), and the frame where it is placed in the tray ($tray_frame$).

Time Slice Cutter crops the user-recorded disassembly video based on the t_{start} and t_{end} . Although it is currently performed

manually, we plan to automate this process using tools such as ffmpeg.

Tray Analyzer extract visual attributes—such as geometry, color, and tray position—from the $tray_frame$ for each component with vision model such as OpenCV.

Component Inspector & Relation Analyzer Agent infer component names, functional descriptions, and inter-component relationships.

Reassembly Planner Agent generates reverse assembly steps to guide reassembly.

Finally, we built the AR interface in Unity and deployed it on Meta Quest 3.

4 Early Evaluation and Ongoing Work

We conducted an early evaluation of our pipeline using disassembly videos of six household devices: a kettle, a handheld fan, a microscope toy, an AC remote and a robot vacuum cleaner. We found that the task descriptions were generally accurate. Among the 32 components, 29 were correctly named.

For future work, we plan to integrate lightweight computer vision models to assist component shape, size, and color recognition, and employ prompt engineering techniques to improve the accuracy of component spatial relationship extraction. To reduce AR overlay occlusion of critical interaction areas and reduce users’ context switching between physical manipulation and virtual guidance, we

also plan to spatially anchor ARoverlays to actual components and adapt their placement based on user context [1].

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